Vector Calculus

Vector Calculus

COORDINATE SYSTEMS

- RECTANGULAR or Cartesian
  - CYLINDRICAL
  - SPHERICAL

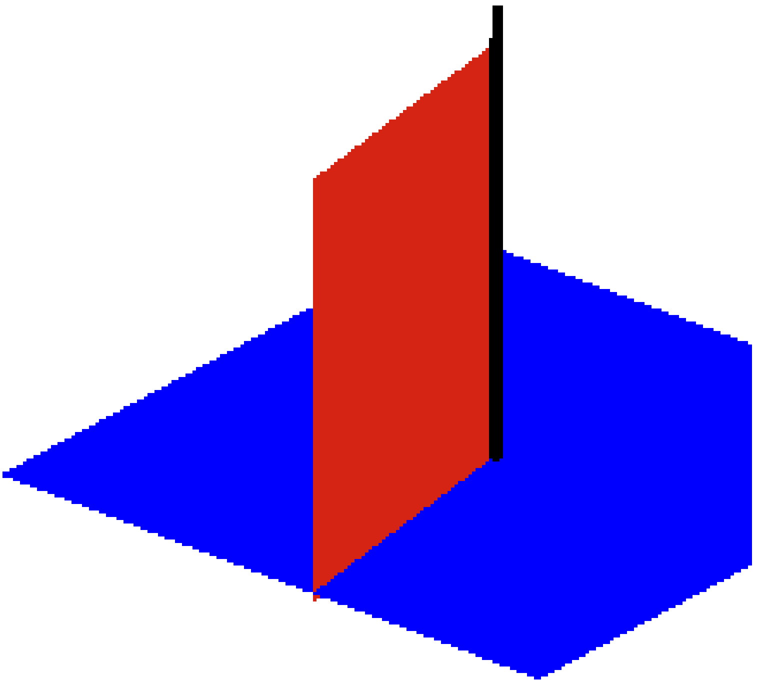
Choice is based on symmetry of problem

<u>Examples:</u>

Sheets - RECTANGULAR

Wires/Cables - CYLINDRICAL

Spheres - SPHERICAL



Cylindrical Symmetry

Spherical Symmetry

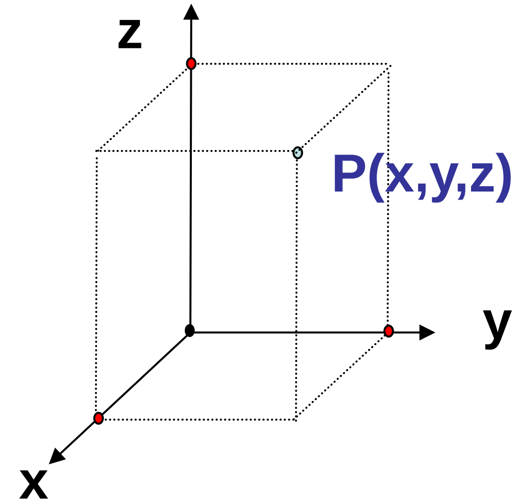
Cartesian Coordinates Or Rectangular Coordinates

P (x, y, z)

$$-\infty < x < \infty$$

$$-\infty < y < \infty$$

$$-\infty < z < \infty$$



A vector A in Cartesian coordinates can be written as

$$(A_x, A_y, A_z) \quad \text{or} \quad A_x a_x + A_y a_y + A_z a_z$$

where a_x,a_y and a_z are unit vectors along x, y and z-directions.

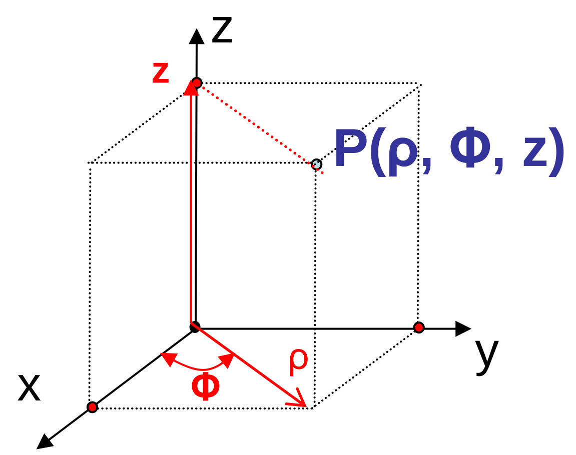
Cylindrical Coordinates

P (ρ, Φ, z)

$$0 \le \rho < \infty$$

$$0 \le \phi < 2\pi$$

$$-\infty < z < \infty$$



A vector A in Cylindrical coordinates can be written as

$$(A_\rho, A_\phi, A_z) \text{ or } A_\rho a_\rho + A_\phi a_\phi + A_z a_z$$

where a_ρ,a_Φ and a_z are unit vectors along ρ, Φ and z-directions.

x= ρ cos Φ, y=ρ sin Φ, z=z

$$\rho = \sqrt{x^2 + y^2}, \phi = \tan^{-1}\frac{y}{x}, z = z$$

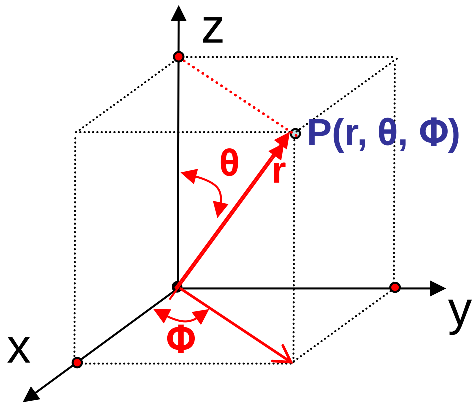
Spherical Coordinates

P (r, θ, Φ)

$$0 \le r < \infty$$

$$0 \le \theta \le \pi$$

$$0 \le \phi < 2\pi$$



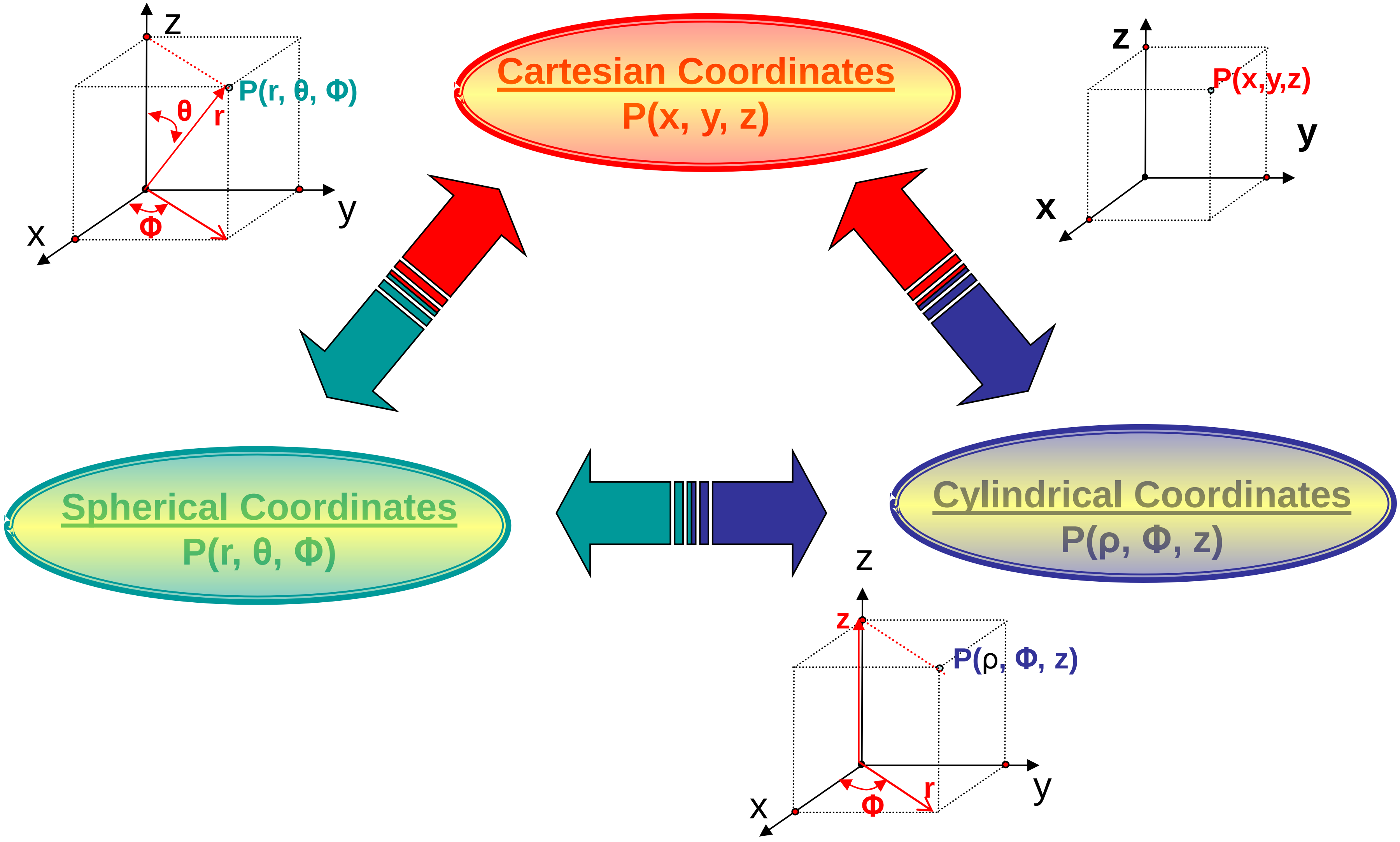
A vector A in Spherical coordinates can be written as

$$(A_r, A_\theta, A_\phi) \quad \text{or} \quad A_r a_r + A_\theta a_\theta + A_\phi a_\phi$$

where a_r, a_θ, and a_Φ are unit vectors along r, θ, and Φ-directions.

x=r sin θ cos Φ, y=r sin θ sin Φ, Z=r cos θ

$$r = \sqrt{x^2 + y^2 + z^2}, \theta = \tan^{-1}\frac{\sqrt{x^2 + y^2}}{z}, \phi = \tan^{-1}\frac{y}{x}$$



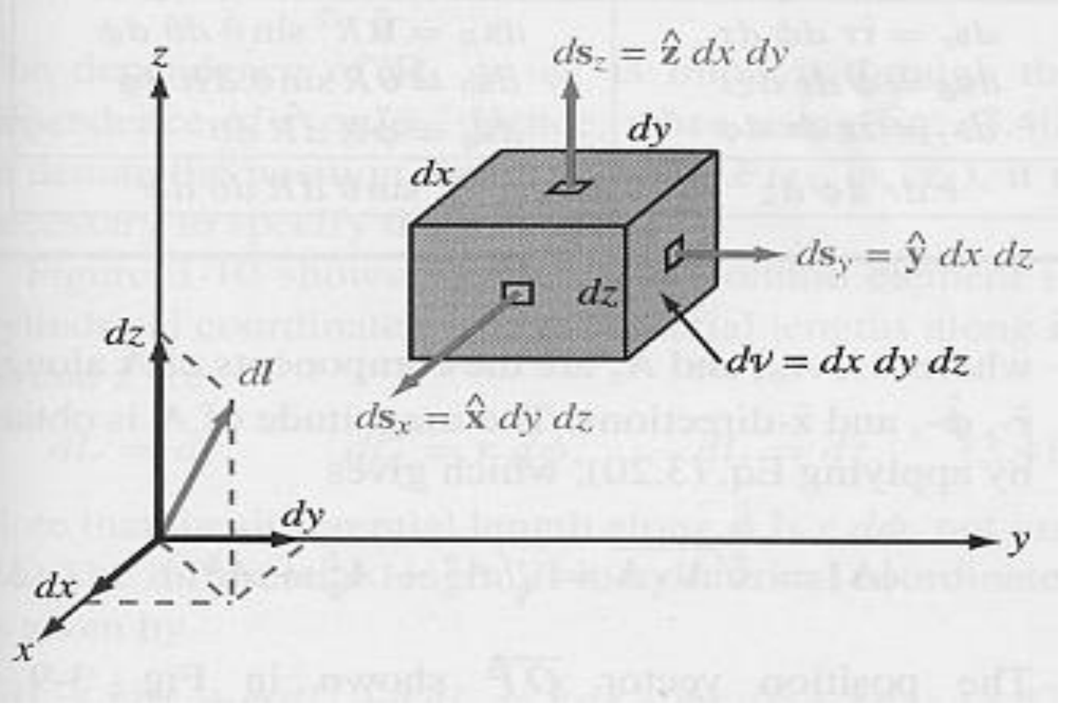
Cartesian Coordinates
P(x, y, z)
Spherical Coordinates
P(r, θ, Φ)
Cylindrical Coordinates
P(ρ, Φ, z)
z
y
x
P(r, θ, Φ)
θ
r
Φ
z
y
x
P(x,y,z)
z
y
x
z
P(ρ, Φ, z)
r
Φ

Differential Length, Area and Volume

Cartesian Coordinates

Differential displacement

$$dl = dx a_x + dy a_y + dz a_z$$



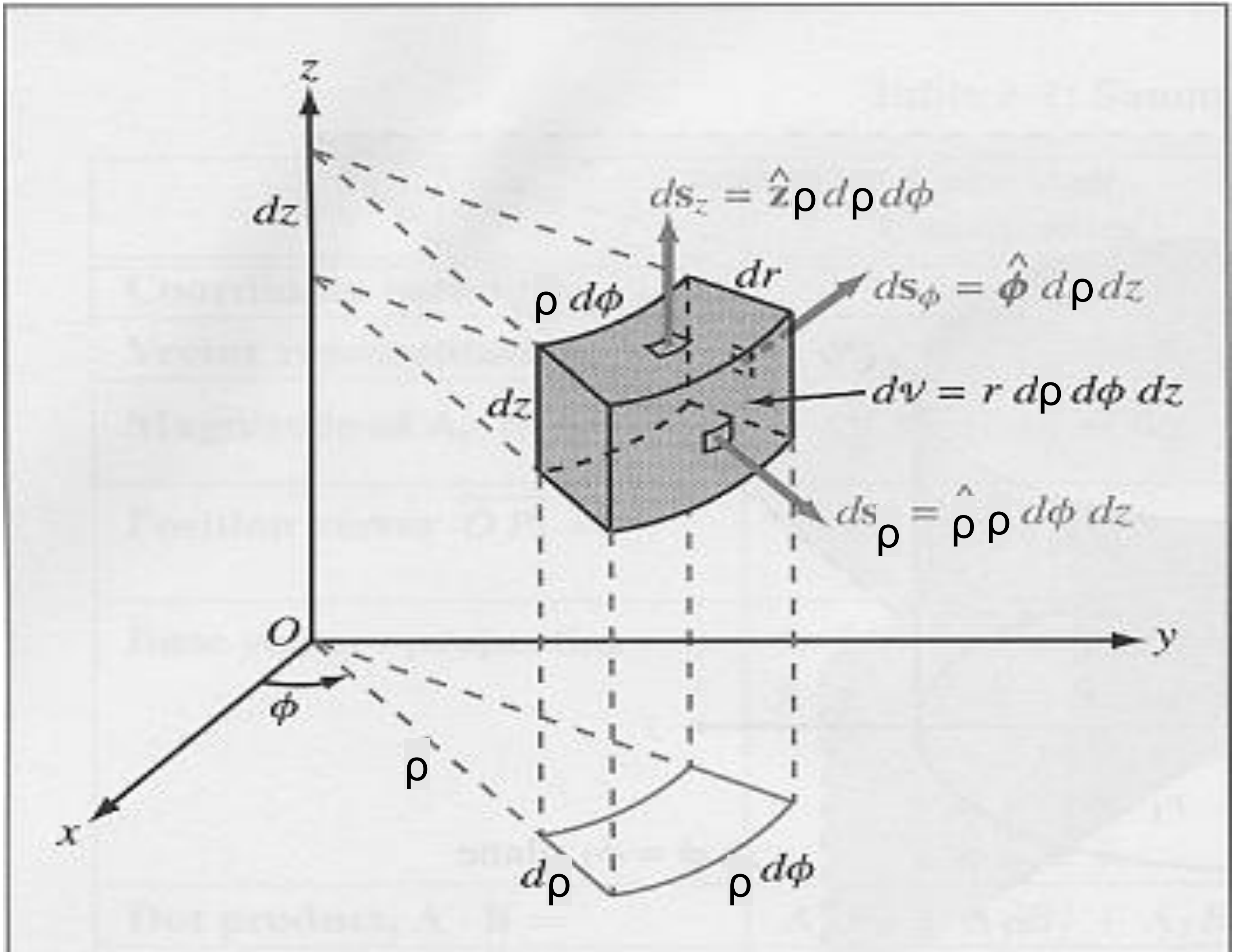
Differential area

$$dS = dydz a_x = dxdz a_y = dxdy a_z$$

Differential Volume

$$dV = dxdydz$$

Cylindrical Coordinates



Differential Length, Area and Volume

Cylindrical Coordinates

Differential displacement

$$dl = d\rho a_\rho + \rho d\phi a_\phi + dz a_z$$

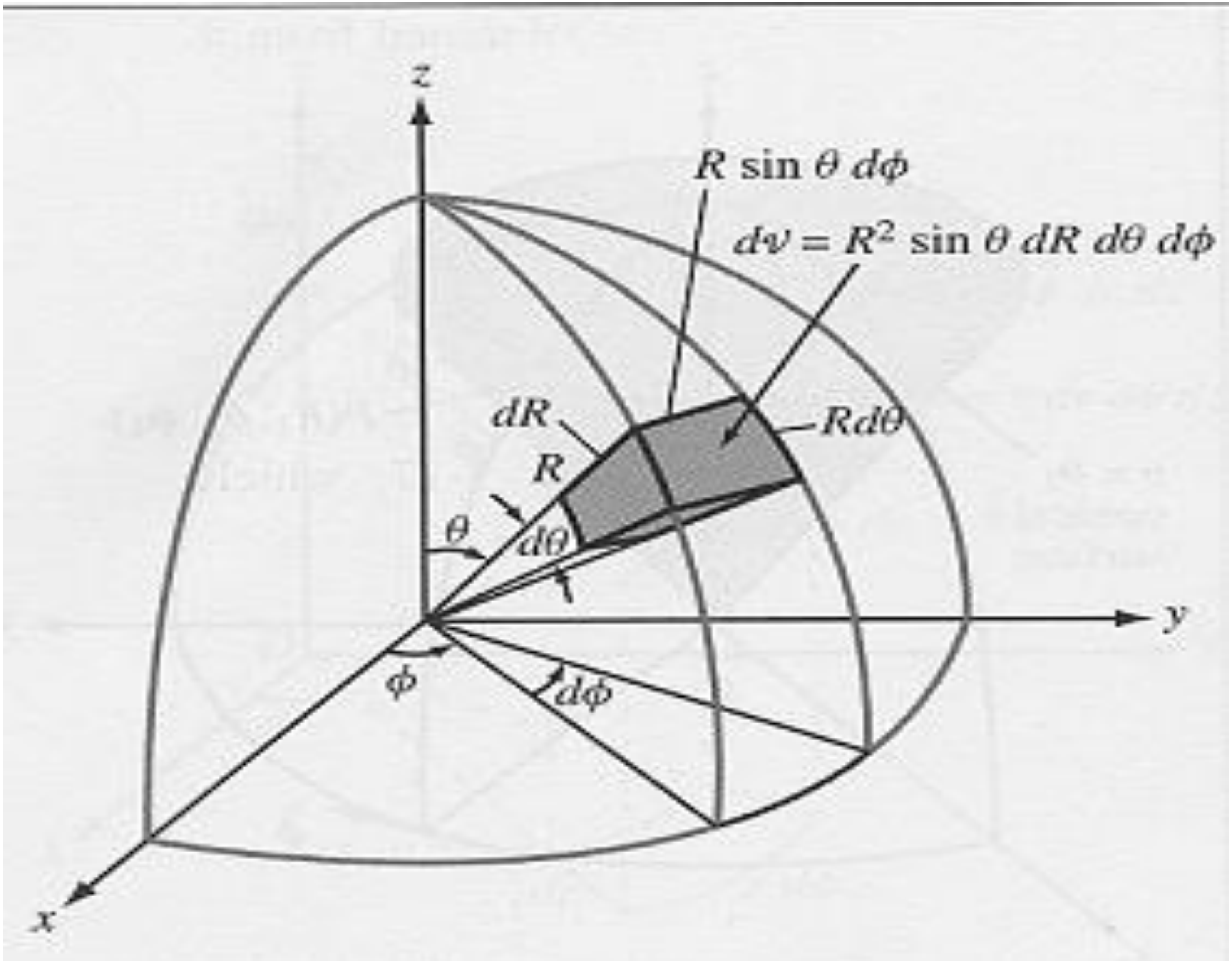
Differential area

$$dS = \rho d\phi dz a_\rho = d\rho dz a_\phi = \rho d\rho d\phi a_z$$

Differential Volume

$$dV = \rho d\rho d\phi dz$$

Spherical Coordinates



Differential Length, Area and Volume

Spherical Coordinates

Differential displacement

$$dl = dr a_r + r d\theta a_\theta + r \sin\theta d\phi a_\phi$$

Differential area

$$dS = r^2 \sin\theta d\theta d\phi a_r = r \sin\theta dr d\phi a_\theta = r dr d\theta a_\phi$$

Differential Volume

$$dV = r^2 \sin\theta dr d\theta d\phi$$

Line, Surface and Volume Integrals

Line Integral

$$\oint_{L} A.dl$$

Surface Integral

$$\psi = \int_{S} A.dS$$

Volume Integral

$$\int_{V} p_{v} dv$$

Gradient, Divergence and Curl

The Del Operator

$$\nabla = \frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k$$

- Gradient of a scalar function is a vector quantity. $\nabla f \longrightarrow$ Vector
- Divergence of a vector is a scalar quantity. $\nabla . A$
- Curl of a vector is a vector quantity. $\nabla \times A$

Del Operator

Cartesian Coordinates

$$\nabla = \frac{\partial}{\partial x} a_x + \frac{\partial}{\partial y} a_y + \frac{\partial}{\partial z} a_z$$

Cylindrical Coordinates

$$\nabla = \frac{\partial}{\partial \rho} a_\rho + \frac{1}{\rho} \frac{\partial}{\partial \phi} a_\phi + \frac{\partial}{\partial z} a_z$$

Spherical Coordinates

$$\nabla = \frac{\partial}{\partial r} a_r + \frac{1}{r} \frac{\partial}{\partial \theta} a_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} a_\phi$$

Gradient of a Scalar

The gradient of a scalar field V is a vector that represents both the magnitude and the direction of the maximum space rate of increase of V.

$$\nabla V = \frac{\partial V}{\partial x} a_x + \frac{\partial V}{\partial y} a_y + \frac{\partial V}{\partial z} a_z$$

$$\nabla V = \frac{\partial V}{\partial \rho} a_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} a_\phi + \frac{\partial V}{\partial z} a_z$$

$$\nabla V = \frac{\partial V}{\partial r} a_r + \frac{1}{r} \frac{\partial V}{\partial \theta} a_\theta + \frac{1}{r \sin\theta} \frac{\partial V}{\partial \phi} a_\phi$$

Gradient of a scalar field

Let $\phi(x, y, z)$ be any continuous differential scalar function. Then, gradient of scalar function $\phi(x, y, z)$ is mathematically written as

$$\vec{\nabla}\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\phi = \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k}$$

$$\text{or } grad\,\phi = \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k}$$

$grad\,\phi$ is a vector.

Physical Significance

$$grad\,\phi = \frac{\partial \phi}{\partial n}\hat{n}$$

Thus gradient of a scalar function ϕ at any point is a vector whose magnitude at any point is equal to the maximum rate of increase of ϕ and directed along the direction in which the maximum change occurs i.e., perpendicular to level surface at that point. *In simple word, gradient of scalar function represent the maximum rate of change of scalar function.*

Divergence of a Vector

The divergence of A at a given point P is the outward flux per unit volume as the volume shrinks about P.

$$divA = \nabla .A = \lim_{\Delta v \to 0} \frac{\oint_S A.dS}{\Delta v}$$

$$\nabla .A = \frac{\partial A}{\partial x} + \frac{\partial A}{\partial y} + \frac{\partial A}{\partial z}$$

$$\nabla .A = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

Curl of a Vector

The curl of A is an axial vector whose magnitude is the maximum circulation of A per unit area tends to zero and whose direction is the normal direction of the area when the area is oriented to make the circulation maximum.

$$curl A = \nabla \times A = \left(\lim_{\Delta s \to 0} \frac{\oint_L A.dl}{\Delta S} \right)_{\max} a_n$$

Where ΔS is the area bounded by the curve L and a_n is the unit vector normal to the surface ΔS

$$\nabla \times A = \begin{bmatrix} a_x & a_y & a_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ A_x & A_y & A_z \end{bmatrix}$$

Cartesian Coordinates

$$\nabla \times A = \frac{1}{\rho} \begin{bmatrix} a_\rho & \rho a_\phi & a_z \\ \dfrac{\partial}{\partial \rho} & \dfrac{\partial}{\partial \phi} & \dfrac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{bmatrix}$$

Cylindrical Coordinates

$$\nabla \times A = \frac{1}{r^2 \sin\theta} \begin{bmatrix} a_r & r a_\theta & r \sin\theta a_\phi \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial \theta} & \dfrac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin\theta A_\phi \end{bmatrix}$$

Spherical Coordinates

Divergence or Gauss' Theorem

The divergence theorem states that the total outward flux of a vector field A through the closed surface S is the same as the volume integral of the divergence of A.

$$\oint A.dS = \int_V \nabla.A dv$$

Stokes' Theorem

Stokes's theorem states that the circulation of a vector field A around a closed path L is equal to the surface integral of the curl of A over the open surface S bounded by L, provided A and $\nabla \times A$ ***are continuous on S***

$$\oint_L A.dl = \int_S (\nabla \times A).dS$$

Q.1 Vector of magnitude of 5 units making an angle 60° with the x-axis. Find its components along x-axis and y-axis.

Horizontal component $A_x = |A| \cos\alpha = 5\cos 60^\circ = 5 \times \frac{1}{2} = 2.5$ units

Vertical component $A_y = |A| \sin\alpha = 5\sin 60^\circ = 5 \times \frac{\sqrt{3}}{2}$ units

Q.2 One of the rectangular components of the force 10 N is 8 N. Find the other components of the force.

F=10 N Let F_x= 8 N and F_y=?

$$F = \sqrt{F_x^{\ 2} + F_y^{\ 2}} \Rightarrow F_y = \sqrt{F^2 - F_x^{\ 2}}$$

$$= \sqrt{(10)^2 - (8)^2} = \sqrt{36} = 6$$

$$F_y = 6\,\text{N}$$

Q.3 $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ **and** $\vec{B} = 3\hat{i} + 4\hat{j} - 9\hat{k}$. **What is the angle between these two vectors** $\vec{A}$ **and** $\vec{B}$

$$\cos\theta = \frac{\vec{A}.\vec{B}}{AB}$$

$$|A| = \sqrt{4+9+4} = \sqrt{17}$$

$$|B| = \sqrt{9+16+81} = \sqrt{106}$$

$$A \cdot B = 6 + 12 - 18 = 0$$

$$\text{Cos}\,\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

Q.4 If $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} + 9\hat{k}$. Find the scalar and cross product of two vectors.

Scalar product= $\vec{A} \cdot \vec{B} = (2\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (3\hat{i} + 4\hat{j} + 9\hat{k}) = 6 + 12 + 18 = 36$

Cross product $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 2 \\ 3 & 4 & 9 \end{vmatrix}$

Find a vector which is perpendicular to $\vec{A} = 6\hat{i} + 9\hat{j}$ and $\vec{B} = -2\hat{i} + 7\hat{j}$

$$\hat{n} = \frac{\vec{A}\times\vec{B}}{\left|\vec{A}\times\vec{B}\right|}$$

$$\vec{A}\times\vec{B} = \hat{i}(0) - \hat{j}(0) + \hat{k}(42+18)$$

$$\vec{A}\times\vec{B} = 60\hat{k}$$

$$\left|\vec{A}\times\vec{B}\right| = 60$$

$$\hat{n} = \frac{\vec{A}\times\vec{B}}{\left|\vec{A}\times\vec{B}\right|} = \frac{60\hat{k}}{60} = \hat{k}$$